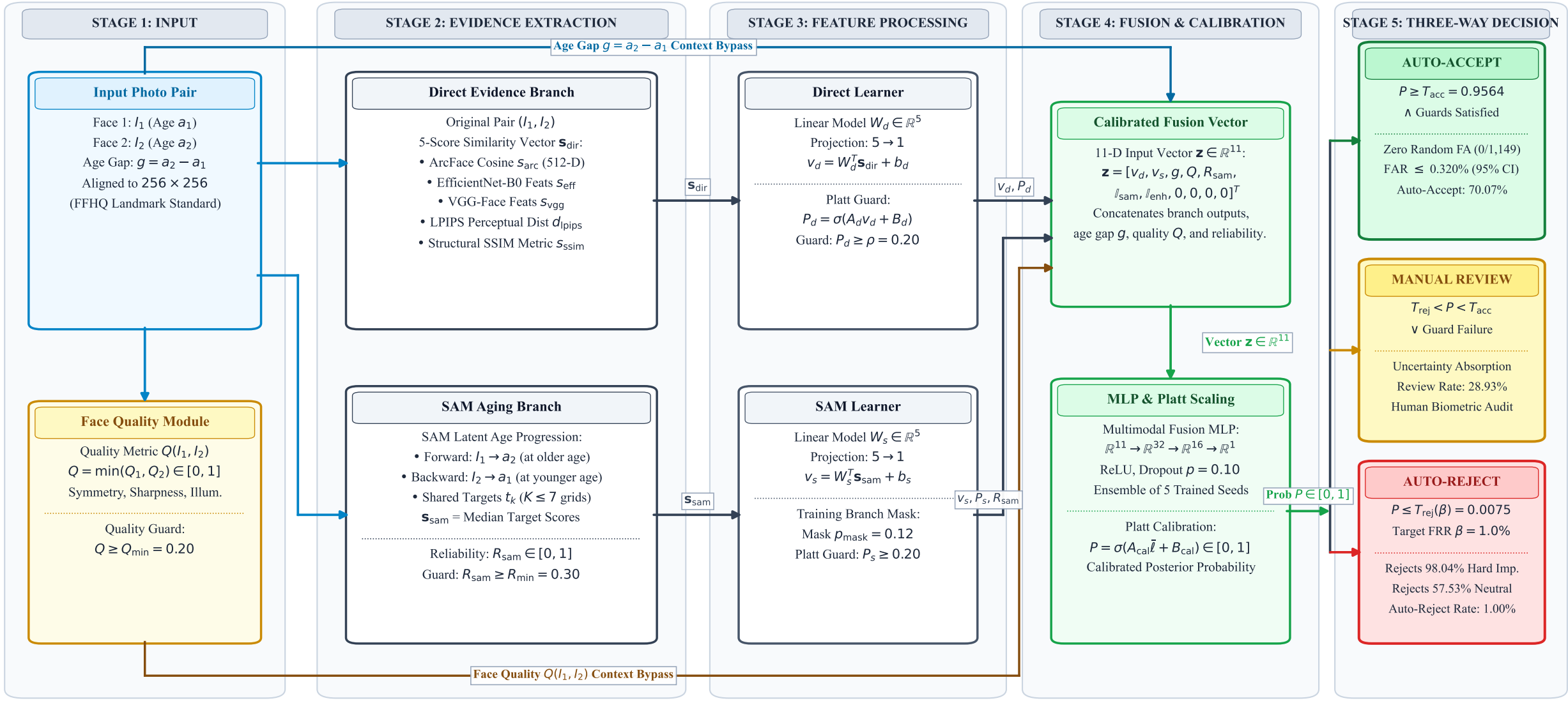

STAGE 1: INPUT
STAGE 2: EVIDENCE EXTRACTION
STAGE 3: FEATURE PROCESSING
STAGE 4: FUSION & CALIBRATION
STAGE 5: THREE-WAY DECISION
Age Gap $g = a_2 - a_1$ Context Bypass
Input Photo Pair
Face 1: I_1 (Age a_1)
Face 2: I_2 (Age a_2)
Age Gap: $g = a_2 - a_1$
Aligned to 256×256
(FFHQ Landmark Standard)
Face Quality Module
Quality Metric $Q(I_1, I_2)$
$Q = \min(Q_1, Q_2) \in [0, 1]$
Symmetry, Sharpness, Illum.
Quality Guard:
$Q \geq Q_{\text{min}} = 0.20$
Direct Evidence Branch
Original Pair (I_1, I_2)
5-Score Similarity Vector $\mathbf{s}_{\text{dir}}$:
• ArcFace Cosine s_{arc} (512-D)
• EfficientNet-B0 Feats s_{eff}
• VGG-Face Feats s_{vgg}
• LPIPS Perceptual Dist d_{lpips}
• Structural SSIM Metric s_{ssim}
SAM Aging Branch
SAM Latent Age Progression:
• Forward: $I_1 \rightarrow a_2$ (at older age)
• Backward: $I_2 \rightarrow a_1$ (at younger age)
• Shared Targets t_k ($K \leq 7$ grids)
$\mathbf{s}_{\text{sam}}$ = Median Target Scores
Reliability: $R_{\text{sam}} \in [0, 1]$
Guard: $R_{\text{sam}} \geq R_{\text{min}} = 0.30$
$\mathbf{s}_{\text{dir}}$
$\mathbf{s}_{\text{sam}}$
Direct Learner
Linear Model $W_d \in \mathbb{R}^5$
Projection: $5 \rightarrow 1$
$v_d = W_d^T \mathbf{s}_{\text{dir}} + b_d$
Platt Guard:
$P_d = \sigma(A_d v_d + B_d)$
Guard: $P_d \geq \rho = 0.20$
SAM Learner
Linear Model $W_s \in \mathbb{R}^5$
Projection: $5 \rightarrow 1$
$v_s = W_s^T \mathbf{s}_{\text{sam}} + b_s$
Training Branch Mask:
Mask $p_{\text{mask}} = 0.12$
Platt Guard: $P_s \geq 0.20$
v_d, P_d
v_s, P_s, R_{sam}
Face Quality $Q(I_1, I_2)$ Context Bypass
Calibrated Fusion Vector
11-D Input Vector $\mathbf{z} \in \mathbb{R}^{11}$:
$\mathbf{z} = [v_d, v_s, g, Q, R_{\text{sam}}, \mathbb{I}_{\text{sam}}, \mathbb{I}_{\text{enh}}, 0, 0, 0]^T$
Concatenates branch outputs, age gap g, quality Q, and reliability.
Vector $\mathbf{z} \in \mathbb{R}^{11}$
MLP & Platt Scaling
Multimodal Fusion MLP:
$\mathbb{R}^{11} \rightarrow \mathbb{R}^{32} \rightarrow \mathbb{R}^{16} \rightarrow \mathbb{R}^1$
ReLU, Dropout $p = 0.10$
Ensemble of 5 Trained Seeds
Platt Calibration:
$P = \sigma(A_{\text{cal}} \bar{\ell} + B_{\text{cal}}) \in [0, 1]$
Calibrated Posterior Probability
Prob $P \in [0, 1]$
AUTO-ACCEPT
$P \geq T_{\text{acc}} = 0.9564$
∧ Guards Satisfied
Zero Random FA (0/1,149)
FAR $\leq$ 0.320% (95% CI)
Auto-Accept: 70.07%
MANUAL REVIEW
$T_{\text{rej}} < P < T_{\text{acc}}$
∨ Guard Failure
Uncertainty Absorption
Review Rate: 28.93%
Human Biometric Audit
AUTO-REJECT
$P \leq T_{\text{rej}}(\beta) = 0.0075$
Target FRR $\beta = 1.0\%$
Rejects 98.04% Hard Imp.
Rejects 57.53% Neutral
Auto-Reject Rate: 1.00%